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Unit 7: Legal and Ethical Aspects of Computing System

A Guardian's Guide to the Digital World

Welcome, Digital Guardian

Every time you open an app, click "I Agree," or scroll through social media, you are stepping into a world that has its own laws, its own dangers, and its own moral questions. Most people move through this world on autopilot — tapping "Accept" without reading, sharing photos without thinking, and clicking links without checking.

You are not going to be one of those people.

By the end of this unit, you will think like a **digital guardian** — someone who understands the rules of the online world well enough to protect themselves, protect others, and even help design a fairer digital future. We'll cover:

- The legal agreements you sign every day without noticing (**Terms of Use**)
- The hidden threats trying to steal your information (**Privacy & Security Threats**)
- Why millions of people are left behind in the digital world (**The Digital Divide**)
- How computing reshapes society for better and worse (**Computing's Impact**)
- How to behave as a responsible, ethical citizen online (**Digital Citizenship**)

One important reminder before we start: **falling for a phishing email or clicking "Agree" without reading is not a sign of being foolish.** These systems are often deliberately designed by professionals to be confusing, rushed, and manipulative. Even experienced engineers and lawyers get tricked sometimes. Your job isn't to be perfect — it's to get better at spotting the patterns. Let's begin.

Introduction: Why Legal and Ethical Literacy Matters

In today's digital world, computing systems play a crucial role in every aspect of our lives — from personal devices to complex networks. As these systems become more deeply woven into our daily routines, understanding the legal and ethical considerations behind them becomes essential, not optional.

This chapter explores terms of use, privacy and security threats, and the digital divide. It covers legal challenges like data protection and intellectual property rights, as well as ethical questions about fairness, privacy, and the broader impact of technology on society.

By the end of this chapter, you will be able to:

- Understand the fundamental legal and ethical considerations in computing systems
 - Identify various privacy and security threats in digital environments
 - Explain strategies for preventing and mitigating security threats
 - Discuss the digital divide and its implications for different social groups
 - Recognize and address bias in computing systems
 - Utilize information safely and responsibly
 - Assess the impact of computing on individuals and society
 - Apply principles of digital citizenship and ethical conduct
-

7.1 Understanding Terms of Use

The Hook (Story Mode)

Imagine you're renting an apartment. You're excited to move in, so you skim the contract, sign quickly, and hand over your deposit. Three months later, you want to leave early — but buried in paragraph 14 of that contract you never fully read, there's an "early termination fee" worth two months' rent. You had the right to know this from day one. You just never read it.

This happens online **constantly**. Every app, every website, every game has its own version of that rental contract — and almost nobody reads it before clicking "I Agree."

The Explanation

"Terms of Use" — also called Terms and Conditions or Terms of Service — are legal agreements that outline how a service or product can be used. These terms are set by the provider (a website, app, or software company) and must be agreed upon by the user. Their main purpose is to create a fair and transparent relationship between the provider and the user, so both sides know their rights and responsibilities.

Example: When you use a Pakistani online shopping platform like Daraz, the Terms of Use may include clauses about payment methods, return policies, and delivery procedures — so customers know exactly what to expect, and what's expected of them.

One-Sentence Summary

Terms of Use are the "rulebook" every digital service asks you to silently accept before you can use it.

Quick Recap

Terms of Use exist to protect both the company and the user by clearly defining the rules of the relationship — even though almost nobody reads them in full.

7.1.1 Importance

The Explanation

Understanding Terms of Use matters for three big reasons:

1. Protection of Rights

Terms of Use protect users by defining their rights — such as the right to privacy, the right to a certain level of service, and the right to seek redress (a fix or compensation) if something goes wrong.

Example: When you use a food delivery service like Foodpanda, the Terms of Use specify what happens if your order arrives wrong or late.

2. Clarity and Transparency

These terms explain exactly what a service will deliver, under what conditions, and what actions aren't allowed — preventing misunderstandings and helping users make informed decisions.

Example: If you use a mobile payment app, the Terms of Use will outline transaction limits, fees, and how to report an unauthorized transaction.

3. Legal Safeguards

For businesses, Terms of Use act as protection against misuse of their services, setting boundaries that shield the company from legal issues caused by users breaking the rules.

Example: A Pakistani online education platform may include terms that prohibit sharing course content without permission, protecting their intellectual property (their original created work).

The Practical Walkthrough: Reading a Terms of Use Like a Pro

Most students skip straight to the "I Agree" button. Here's a fast way to actually extract the important information in under 2 minutes.

Step 1: Use **Ctrl+F** (or **Cmd+F** on Mac) to search the document for key words: "data," "share," "cancel," "refund," "liable," "terminate."

Step 2: Read only the sentences surrounding those key words — skip the rest.

Step 3: Ask yourself three questions:

- What am I giving up? (data, content ownership, rights)
- What am I getting? (service, features, guarantees)
- How do I get out if I want to? (cancellation, deletion of my data)

What just happened? You just turned a 20-page legal wall of text into a 2-minute risk check — a skill professional analysts use every day.

Pause & Think

A free photo-editing app's Terms of Use state that it "owns all photos uploaded and may use them to train AI models." Is this a fair trade for a free service? What exactly is the user giving up, and is it worth what they're getting in return?

Quick Recap

Understanding Terms of Use protects your rights, gives you clarity about what a service actually promises, and shields businesses from misuse — all three exist to make the relationship fair for everyone.

7.1.2 Common Clauses and Conditions

The Explanation

Terms of Use usually contain several common clauses (specific sections), each designed to protect both the provider and the user:

1. User Obligations — What is expected from the user.

Example: Ride-sharing apps like Careem require a valid phone number and prohibit misusing the service for non-transportation purposes.

2. Limitations of Liability — Limits on what the provider is responsible for if something goes wrong.

Example: If an online banking app goes down temporarily, the bank's Terms of Use typically limit their liability for the disruption.

3. Privacy and Data Use — How a company collects, uses, and protects user data.

Example: WhatsApp's Terms of Use explain how your messages are stored and protected.

4. Intellectual Property Rights — Protects content owned by the provider, like logos, software, and proprietary materials.

Example: A Pakistani news website may prohibit copying and redistributing its articles without permission.

5. Termination of Service — The conditions under which a provider can end a user's access.

Example: Facebook's terms allow suspending accounts that violate community standards, such as spreading hate speech or misinformation.

Clause Quick-Reference Table

Clause	Protects	Real Example
User Obligations	The provider, from misuse	Careem requiring a valid phone number
Limitations of Liability	The provider, from lawsuits over outages	Bank app downtime disclaimers
Privacy and Data Use	The user, regarding personal data	WhatsApp message storage policy
Intellectual Property Rights	The provider's content	News site article copying ban
Termination of Service	The platform's community	Facebook account suspension rules

Grab a Partner

Partner A picks an app both of you use daily (e.g., TikTok, WhatsApp, a mobile game). Partner B searches its Terms of Use for one clause matching each category above. Compare notes: which clause was hardest to find, and why do you think it was written that way?

Quick Recap

Every Terms of Use document is built from a predictable set of clauses — obligations, liability limits, privacy rules, intellectual property protection, and termination conditions — and learning to spot them quickly makes any contract far less intimidating.

7.1.3 Legal Ramifications

The Explanation

Ramifications simply means "consequences." Violating Terms of Use can lead to serious legal consequences: users can be fined, sued, or banned from a service entirely.

Example: Using software without a proper license may result in legal action for software piracy. Companies using unlicensed software face both legal risk and cybersecurity threats, since pirated software often lacks proper security updates.

Pause & Think

A student downloads a "cracked" (illegally unlocked) version of expensive design software to avoid paying for it. Beyond the legal risk of piracy, what *cybersecurity* risk might this student also be unknowingly accepting? (Hint: where did the "crack" file actually come from, and who modified it?)

Quick Recap

Breaking Terms of Use isn't just against the rules — it can carry real legal and financial consequences, and often introduces security risks that aren't obvious at first glance.

7.1.4 Ethical Considerations

The Explanation

Ethical considerations in Terms of Use include **fairness**, **transparency**, and **respect for user rights**. Companies have an ethical duty to make their terms clear, straightforward, and not deceptive.

Example: When users sign up for an online course in Pakistan, the terms should clearly state whether a certificate will be awarded, how personal data will be handled, and whether hidden fees exist.

Pause & Think

A free photo-editing app's Terms of Use state that it "owns all photos uploaded and may use them to train AI models." Revisit this scenario from 7.1.1 — but now specifically through an **ethics** lens: even if this clause is technically legal, is it ethical? What rights is the user surrendering that they may not fully realize?

Quick Recap

Legal and ethical are not always the same thing — a clause can be perfectly legal while still being unfair or deceptive, which is exactly why ethical scrutiny matters even after checking the law.

7.1.5 Personal Rights

The Explanation

Personal rights within Terms of Use include the **right to privacy**, the **right to be informed**, and the **right to withdraw consent** (to change your mind and stop a company from using your data).

Example: Under Pakistan's Personal Data Protection Bill, users have the right to access their personal data and request corrections or deletions. When using local e-commerce websites, users have the right to know how their personal information — like addresses and phone numbers — will be used.

Do You Know?

Some online platforms frequently update their Terms of Use. It's good practice to review them periodically to stay informed about any changes.

Quick Recap

You have real, defined rights over your own data and privacy — knowing them is the first step to actually being able to use them.

7.2 Privacy and Security Threats

The Hook (Story Mode)

In 2013, retail giant Target suffered one of the most infamous data breaches in history. Attackers didn't break into Target's main systems directly — instead, they stole login credentials from a small third-party **heating and air conditioning vendor** that had remote access to Target's network for maintenance purposes. That one small, overlooked vulnerability gave attackers a path straight into Target's payment systems, exposing over **40 million credit and debit cards**.

The lesson? In digital security, the weakest link in the chain — no matter how small or unrelated it seems — can bring down an entire system. This is why understanding every type of security threat matters, even the ones that seem minor at first glance.

The Explanation

In today's digital world, privacy and security threats are major concerns for individuals and organizations alike. These threats can compromise personal information, financial data, and overall online security.

7.2.1 Types of Security Threats

7.2.1.1 Spam

The Explanation

Spam refers to unwanted messages you receive by email or phone.

Example: You may receive numerous text messages from unknown numbers offering products or services you never requested.

Quick Recap

Spam is unwanted, unsolicited digital messaging — annoying, but usually the least dangerous entry on this list.

7.2.1.2 Spyware

The Hook (Story Mode)

Imagine someone quietly sitting in the backseat of your car with a notepad, silently recording every store you visit, every friend you pick up, and every conversation you have — then selling that notebook to advertisers without you ever knowing they were there. That is exactly what spyware does to your digital life.

The Explanation

Spyware is harmful software that secretly monitors your activities on your computer or mobile phone.

Example: If you download a free app from an unreliable source, it may install spyware on your device. This spyware can secretly monitor your online activities — like websites you visit or information you type — without your knowledge.

Pause & Think

A "free flashlight app" requests permission to access your contacts, microphone, and location — none of which a flashlight actually needs to function. What should this mismatch immediately signal to you?

Quick Recap

Spyware is software that spies on you silently from inside your own device — the biggest warning sign is an app requesting far more access than its actual function requires.

7.2.1.3 Cookies

The Hook (Story Mode)

Think of a cloakroom ticket at a theater, or a stamp on your hand at an event. It's a small token given to you so the system remembers who you are when you walk back through the door — you don't have to re-introduce yourself every single time.

The Explanation

Cookies are small files that websites place on your device when you visit them. They help a website remember things like your login details and personal preferences.

Example: When you visit an online shopping site, cookies save your login information so you don't need to re-enter it every visit. However, cookies can also track your online activity — like the products you view — to show you related advertisements later.

First-Party vs. Third-Party Cookies

Feature	First-Party Cookies	Third-Party Cookies
Set by	The website you're directly visiting	A different domain (usually an advertiser) embedded in that site
Purpose	Remembering login, cart items, language settings	Tracking your behavior across multiple different websites
Privacy Risk	Low — stays within one site's context	Higher — builds a profile of you across the entire internet
Common Use	"Stay logged in" checkbox	Ads that "follow you" from site to site

The Practical Walkthrough: Managing Cookies in Your Browser

Step 1: Open your browser's Settings menu (Chrome, Firefox, or Edge).

Step 2: Navigate to **Privacy and Security** → **Cookies and Site Data**.

Step 3: Look for an option like "Block third-party cookies" and enable it.

Step 4: To clear existing cookies, find "Clear browsing data," select "Cookies and other site data," and choose a time range (e.g., "All time").

Step 5: Click "Clear Data."

What just happened? You just told your browser to stop letting outside advertisers track your movement across different sites, while still allowing individual websites to remember your login for convenience.

Grab a Partner

Partner A argues cookies should always be fully blocked for maximum privacy. Partner B argues some cookies (like login-session cookies) are actually essential for a usable internet. Debate: is there a middle ground? What would you personally choose to block, and what would you choose to allow?

Quick Recap

Cookies are small memory files that make browsing convenient, but first-party and third-party cookies carry very different privacy risks — and modern browsers let you control both.

7.2.1.4 Phishing

The Explanation

Phishing is a scam where someone pretends to be a trustworthy organization to trick you into giving away personal information.

Example: You may receive an email that looks like it's from your bank, asking you to click a link and enter your account details. If you follow these instructions, your money could be at risk — because the email is fake.

Do You Know?

Over 90% of phishing attacks are carried out through email. These scams often look like legitimate communications from trusted organizations, making it crucial to verify the source before clicking any links or providing personal information.

Quick Recap

Phishing tricks a human into voluntarily handing over sensitive information by impersonating a trusted source — the deception happens through *disguise*, not through hacking.

Pharming: A Related but Different Threat

The Hook (Story Mode)

In 2007, attackers demonstrated a technique even sneakier than phishing: they poisoned **DNS servers** — think of DNS as the internet's phonebook, translating a website name like "mybank.com" into the numerical address your computer actually uses to find it. By corrupting this "phonebook," attackers could secretly redirect people to fake banking sites *without the victim ever mistyping a single letter of the URL*. Basic caution — like double-checking the spelling of a web address — became completely useless against this attack.

The Explanation

Pharming is a sophisticated technique where users are redirected to a fake website without their knowledge — even when they typed the correct, legitimate address.

Example: Imagine you want to access your online banking account, but due to a pharming attack, you're sent to a counterfeit site that looks exactly like your bank's real site. If you enter your login information there, the scammer collects your details.

Phishing vs. Pharming — Side by Side

Feature	Phishing	Pharming
How it works	A fake email/message with a deceptive link	The internet's own address system (DNS) is hijacked
Requires victim to...	Click a link and enter details	Nothing extra — even a correctly typed URL can be redirected
How to spot it	Check sender address, hover over links, look for urgency/pressure	Much harder — check for "https://" and valid security certificate on the site itself
Difficulty to detect	Moderate — visible red flags exist	High — the deception is invisible to the user

One-sentence summary: *Phishing tricks you into handing over your credentials on a fake page, while pharming secretly hijacks the internet's phonebook to send you to a fake site automatically.*

Quick Recap

Phishing relies on tricking a human with a deceptive message; pharming bypasses the human entirely by corrupting the internet's underlying address system — which is exactly why pharming is so much harder to detect.

7.2.2 Security Threat Prevention Techniques

7.2.2.1 Spam Filters

Spam filters help keep unwanted emails out of your inbox.

Example: If you receive a lot of unwanted promotional emails, a spam filter can automatically move them to a separate folder or delete them, so you only see the emails you actually want.

Quick Recap

Spam filters automatically sort unwanted messages away from your main inbox, reducing clutter and risk.

7.2.2.2 Spyware Protection (Anti-Malware & Permissions)

The Explanation

Antivirus/anti-malware software protects your computer from harmful programs like spyware by scanning files for hidden threats before you open them.

Example: If you download a file from the internet, antivirus software checks it for hidden threats. If the file is safe, you can open it worry-free. If it's harmful, the software alerts you and blocks it.

The Practical Walkthrough: Auditing App Permissions

Step 1: Open your phone's Settings app.

Step 2: Navigate to **Apps** → **App Permissions** (or **Privacy** on some devices).

Step 3: Select a permission category — e.g., "Microphone," "Location," or "Contacts."

Step 4: Review the full list of apps that currently have access to that permission.

Step 5: Ask yourself: *"Does this app actually need this to function?"* A flashlight app doesn't need your contacts. A messaging app doesn't need your location unless it has a "share location" feature you actually use.

Step 6: Toggle off (revoke) access for any app whose permission doesn't match its actual purpose.

What just happened? You just closed a door that spyware-like behavior could have used to quietly collect data you never intended to share.

Before / After Permission Audit Example

App	Permission Requested	Actually Needed?	Action
Flashlight App	Contacts, Microphone, Location	No	Revoke all three
Messaging App	Contacts, Microphone (for voice notes)	Yes	Keep both
Weather App	Location	Yes (for local forecast)	Keep
Weather App	Contacts	No	Revoke

Pause & Think

You revoke a permission and an app suddenly stops working correctly. What does this suggest about whether that permission was actually necessary — and what should you do next: restore the permission, or find an alternative app?

Quick Recap

Anti-malware software catches hidden threats in files, while regularly auditing app permissions closes off the unnecessary access points spyware-like behavior depends on.

7.2.2.3 Cookie Management

Refer to the full walkthrough in **7.2.1.3** above — managing cookies means deciding which ones to keep (like your favorite shopping site, for convenience) and which to delete (unfamiliar or unused sites, for privacy).

Example: You can keep cookies from your favorite shopping website so you don't have to log in every time, but it's a good idea to delete cookies from unfamiliar sites to protect your privacy.

Quick Recap

Selective cookie management balances convenience against privacy risk — you don't need to block everything, just the sources you don't trust.

7.2.2.4 Recognizing Phishing

The Practical Walkthrough: Anatomy of a Phishing Email

Let's break down a suspicious email step by step.

```
From: "Your Bank Security" <support@yourbank-secure-verify.com>
Subject: URGENT: Your account will be suspended in 24 hours!

Dear Customer,

We detected unusual activity. Click below immediately to verify
your account or it will be permanently locked.

[Verify My Account Now] → hxyp://yourbank.secure-login-update.ru/verify
```

Step 1: Check the sender's domain. A real bank sends email from its actual domain (e.g., `@yourbank.com`), not a look-alike like `yourbank-secure-verify.com`.

Step 2: Look for urgency and fear. Phrases like "24 hours" or "permanently locked" are designed to make you panic and act before thinking.

Step 3: Hover over the link (don't click!). The visible text says "yourbank.com," but the actual destination is a completely different, suspicious domain ending in `.ru`.

Step 4: Check the greeting. Real banks usually use your actual name ("Dear Ali Ameer"), not a

generic "Dear Customer."

Step 5: Verify independently. Instead of clicking the link, open your browser separately and type your bank's known, correct web address directly, or call their official support number.

What just happened? You just identified four separate red flags — a spoofed domain, manufactured urgency, a mismatched link, and a generic greeting — without clicking a single dangerous link.

Quick Recap

Recognizing phishing means slowing down and checking the sender's real domain, the actual link destination, and the emotional pressure tactics being used — the deception almost always leaves visible clues if you look closely.

7.2.2.5 Guarding Against Pharming

The Explanation

To guard against pharming, always check that a website's address is correct and secure — look for "**https://**" (the "s" means secure) and a **padlock symbol** in the browser's address bar.

The Practical Walkthrough

Step 1: Before entering any login details, glance at the address bar.

Step 2: Confirm the URL begins with **https://**, not just **http://**.

Step 3: Click the padlock icon to view the site's security certificate details — confirm it matches the organization you expect (e.g., "Issued to: YourBank Inc.").

Step 4: If anything looks unusual — a certificate warning, a mismatched organization name, or a missing padlock — stop and do not enter any information.

Grab a Partner

Partner A creates a fake bank email scenario using **phishing** (a deceptive link in a message). Partner B describes a scenario using **pharming** (DNS redirection, even with a correctly typed URL). Discuss: why is pharming so much harder for a regular user to spot than phishing?

Quick Recap

Guarding against pharming means checking a website's actual security certificate and connection type, not just its visible address — since the address itself can be silently hijacked.

7.3 The Digital Divide and Its Impacts

The Hook (Story Mode)

When schools around the world suddenly shifted to remote learning during global health emergencies, something became painfully clear: millions of students didn't have reliable household broadband or a dedicated personal device. Overnight, an internet connection stopped being a "nice extra" and became a basic requirement for continuing your education. This crisis didn't create the digital divide — it simply made a hidden inequality impossible to ignore.

The Explanation

The **digital divide** refers to the gap between individuals, communities, and regions that have access to modern Information and Communication Technology (ICT) and those that do not. It's a critical issue because it can lead to unequal opportunities in education, employment, healthcare, and social participation.

7.3.1 Causes of the Digital Divide

1. **Economic Barriers** — Many families cannot afford computers, smartphones, or broadband internet. This is especially visible in developing countries, rural areas, and low-income populations.
2. **Geographical Barriers** — Access to ICT varies sharply between urban and rural areas. In many remote regions, the infrastructure needed for internet connectivity is underdeveloped or missing entirely.
3. **Educational Barriers** — Lack of digital literacy (the skill of knowing *how* to use technology effectively) means that even when technology is available, people may not know how to use it well.
4. **Social Barriers** — Age, gender, and disability can affect access. Older adults may be less inclined to adopt new technology, and in some regions, women face cultural barriers limiting their access to ICT.

Quick Recap

The digital divide isn't caused by one single factor — it's a combination of economic, geographic, educational, and social barriers that compound on top of each other.

7.3.2 Impacts of the Digital Divide

7.3.2.1 Educational Inequality

Access to the internet and digital tools is increasingly essential for education. Students who lack these resources face a significant disadvantage.

Example: During the COVID-19 pandemic, when schools in Pakistan and around the world shifted to online learning, students from low-income families or rural areas struggled to keep up due to lack of internet access or devices — widening the educational gap between socio-economic groups.

Pause & Think

If a government makes all public services, tax filings, and job applications 100% online, what happens to citizens without reliable internet or digital literacy? How can society prevent this kind of exclusion?

Quick Recap

When education moves online without ensuring universal access first, existing inequality gets amplified rather than solved.

7.3.2.2 Economic Disparities

Many jobs today require at least basic digital skills. Those without access to technology or the internet may struggle to secure employment, leading to higher unemployment in already disadvantaged communities.

Example: In Pakistan, access to online job portals and digital banking services is often concentrated in urban areas, widening the economic gap between urban and rural populations.

Quick Recap

Digital access has become directly tied to economic opportunity — being offline increasingly means being locked out of the modern job market.

7.3.2.3 Social and Civic Participation

Access to social media, online news, and digital government services allows people to stay informed and engaged civically. Those without access are often excluded from these opportunities.

Example: In Pakistan, e-government services are more readily available in cities, leaving rural populations with limited means to participate in governmental processes or access social services.

Quick Recap

Civic participation increasingly happens online — meaning being offline can mean being politically and socially invisible.

7.3.2.4 Health Disparities

Digital platforms provide valuable resources for health education, telemedicine, and appointment scheduling. Rural communities with limited internet access may struggle to find reliable health information or join telemedicine consultations.

Quick Recap

Health access is increasingly digital-first, disadvantaging communities without reliable connectivity.

7.3.2.5 Digital Literacy Gap

A significant digital divide also creates a gap in digital *literacy* itself — people without access to technology don't get the chance to develop essential digital skills, reinforcing the divide between tech-savvy and non-tech-savvy populations over time.

Quick Recap

The digital divide isn't just about access to hardware — it perpetuates itself by denying people the chance to build digital skills in the first place.

7.3.3 Bridging the Digital Divide

Bridging the digital divide means making sure everyone has equal access to technology and the internet, giving everyone the same chances to learn and grow.

1. Government Initiatives — Building the infrastructure needed for internet access in underserved areas.

2. Educational Programs — Schools and community centers teaching people how to use

technology.

3. Public-Private Partnerships — Government, businesses, and nonprofits teaming up to make technology more affordable.

Do You Know?

The Government of Pakistan has launched several initiatives to increase internet access in rural areas, including the introduction of mobile broadband services in remote regions.

Class Activity

Think about your own community. Are there people who may not have access to the internet or digital devices? What can be done to help them? Write a short paragraph on how your school or community can help bridge the digital divide.

Quick Recap

Bridging the digital divide requires coordinated action from government, education, and business — no single actor can solve it alone.

7.4 Computing's Impact on Individuals and Society

The Hook (Story Mode)

Computing systems have revolutionized how we live, work, and communicate. The widespread adoption of the internet, smartphones, and Artificial Intelligence has brought both remarkable benefits and serious risks — often at the exact same time.

7.4.1 Positive Impacts

Increased Accessibility to Information — The internet lets people access vast amounts of information on nearly any topic, breaking down barriers to education.

Example: Online educational platforms like Coursera and edX let students from remote areas take courses from prestigious universities worldwide.

Improved Communication — Social media has made it faster and easier to connect globally, and even enabled major social movements like #MeToo to gain global traction. In Pakistan, social media played a crucial role in raising awareness about the 2014 Peshawar school attack, mobilizing support and solidarity.

Economic Growth — Computing has created new industries and jobs. The rise of e-commerce platforms like Daraz has transformed traditional business models, allowing small Pakistani businesses to reach customers nationwide and internationally.

7.4.2 Negative Impacts

The Digital Divide — As covered in 7.3, a significant portion of the population lacks access to modern computing technologies, leading to unequal opportunities.

Misinformation and Fake News — False information can spread rapidly, causing social unrest.

Example: During the COVID-19 pandemic, misinformation about the virus and vaccines spread rapidly on social media, causing confusion and fear. This highlights the need for digital literacy, so people can critically evaluate what they encounter online.

Privacy Concerns — The collection and storage of vast amounts of personal data, without proper regulation, can lead to misuse.

Example: Incidents like the leakage of citizens' data from Pakistan's National Database and Registration Authority (NADRA) underscore the importance of robust data protection measures.

Positive vs. Negative Impacts — Side by Side

Category	Positive Impact	Negative Impact
Information	Global access to education and knowledge	Rapid spread of misinformation
Communication	Global connection, social movements	Cyberbullying, echo chambers
Economy	New industries, job creation, e-commerce growth	Job displacement in some traditional sectors
Data	Personalized services, convenience	Privacy breaches, data misuse
Access	Democratizes knowledge and opportunity	Digital divide leaves many people excluded

Do You Know?

Computing technology has enabled "Internet in a Box" projects that provide offline educational resources. In Pakistan, such initiatives have brought digital learning to remote areas without internet access, demonstrating how technology can bridge educational gaps even in underserved regions.

Pause & Think

The same technology (social media) both raised global awareness after the Peshawar attack *and* spread dangerous misinformation during COVID-19. Does this mean the technology itself is "good" or "bad"? Or does something else determine the outcome?

Quick Recap

Computing's impact on society is rarely one-sided — the very same tool that empowers education, connection, and economic growth can also spread misinformation and violate privacy, depending entirely on how it's designed, regulated, and used.

7.5 Digital Citizenship and Ethical Considerations

The Hook (Story Mode)

In 1808, an Italian inventor named **Pellegrino Turri** built one of the world's first working typewriters — not for offices or businesses, but so his blind friend, Countess Carolina Fantoni da Fivizzano, could write private letters independently. What began as an accessibility solution for one person became a foundational invention used by the entire world for the next two centuries.

This is the heart of digital citizenship and ethics: designing and behaving with everyone in mind — not just the "average" user — often produces innovations and habits that benefit *everyone*.

The Explanation

Digital citizenship involves understanding how to behave responsibly and ethically when using digital technologies and the internet. It covers behaviors and practices that help ensure a positive, secure online experience for everyone.

Example: In Pakistan, students should learn to communicate respectfully in online forums and on social media — avoiding bullying or harassment, not spreading false information, and understanding the importance of digital privacy and account security.

7.5.1 Responsible Digital Behavior

Responsible digital behavior means using technology and the internet in ways that are respectful and safe:

1. Using Strong Passwords — Create passwords that are difficult for others to guess.

Example: Instead of "password123," use a combination of letters, numbers, and symbols.

2. Avoiding Phishing Scams — Don't click suspicious links or open emails from unknown sources asking for personal information.

Example: If you receive an email claiming to be from a bank asking for account details, verify its authenticity before responding.

Quick Recap

Responsible digital behavior starts with simple, consistent habits — strong passwords and healthy skepticism toward unsolicited requests for information.

7.5.2 Ethical Use of Information

7.5.2.1 Responsible Data Sharing

Responsible data sharing means only sharing personal or sensitive information when absolutely necessary, and only with trusted entities.

Example: When applying for a school scholarship online in Pakistan, students should provide personal details only on secure and verified websites, to prevent misuse of their information.

7.5.2.2 Ethical Issues (Plagiarism, Copyright, Data Bias)

1. Respecting Copyright — Do not copy or use someone else's work without permission.

Example: When writing a research paper, students should not copy text from online articles without citing the source.

2. Avoiding Plagiarism — Properly attribute (credit) ideas and information to their original authors. This means citing the source of any quotes or ideas used in assignments.

3. Data Bias — When data used to build systems (like AI models) reflects existing human prejudices, the system can learn and repeat those same unfair patterns, even without anyone intending it.

Pause & Think

A student copies three sentences from a website into their assignment, changes a few words, and doesn't cite the source. Is this plagiarism, even though the wording is "different" now? What would make this acceptable instead?

Quick Recap

Ethical use of information means respecting where ideas and data come from — through proper sharing practices, correct attribution, and awareness that even "neutral" data can carry hidden bias.

7.5.3 Cybersecurity Awareness

Cybersecurity awareness involves understanding and implementing practices to protect yourself from online threats — combining everything learned in Section 7.2 (spam filters, strong passwords, phishing recognition, and cookie/permission management) into a consistent daily habit.

Do You Know?

A cybercrime can be reported to Pakistan's National Response Centre for Cyber Crime (NR3C-FIA), available at <https://www.nr3c.gov.pk>.

Quick Recap

Cybersecurity awareness isn't a single skill — it's the ongoing habit of applying everything you've learned about threats and prevention, every single time you go online.

7.5.4 Collaborative Problem Solving

7.5.4.1 Human-Machine Collaboration

Human-machine collaboration means working *with* technology to solve problems more effectively:

1. Using Tools — Leverage software to assist with tasks.

Example: Students might use spreadsheet software to analyze data for a group project.

2. Combining Skills — Integrate human judgment with machine capabilities for better results.

Example: Using data analysis tools to interpret complex data sets, while applying human judgment to make the final decision.

7.5.4.2 Teamwork and Knowledge Sharing

1. Group Projects — Work together, each contributing your expertise.

Example: In a group science project, students collaborate and share research findings to complete the assignment.

2. Sharing Insights — Exchange ideas with peers to enhance learning.

Example: Participating in study groups or discussions to broaden understanding of a subject.

Pause & Think

A group project uses an AI tool to summarize research articles, but one team member blindly copies the AI's summary without checking it for accuracy. What role should "human judgment" play even when a powerful tool is doing much of the work?

Quick Recap

The best results come from combining what machines do well (processing large amounts of data quickly) with what humans do well (judgment, context, and ethical reasoning) — neither replaces the other.

7.5.5 Creating Accessible Digital Content

The Hook (Story Mode)

Remember Pellegrino Turri's typewriter from the opening of this section — built for one blind friend, but eventually used by the entire world. This is the pattern behind accessible design: building a public ramp alongside stairs doesn't just help wheelchair users — it also helps parents pushing strollers, travelers rolling suitcases, and delivery workers moving heavy boxes.

Accessible design almost always benefits everyone, not just the group it was originally built for.

The Explanation

Creating accessible digital content means designing and producing content that everyone can use, including people with disabilities.

7.5.5.1 Design for Accessibility (ALT Text, Contrast, Screen Readers)

1. Readable Fonts — Use clear, large fonts that are easy to read.

Example: Choose fonts like Arial or Times New Roman and avoid overly decorative styles.

2. Alternative Text (ALT Text) for Images — Provide written descriptions for images so visually impaired users, relying on screen-reading software, can understand the content.

Example: Include descriptive text for charts or graphs in a report — e.g., `alt="Bar chart showing sales rising from January to March"` instead of leaving it blank.

3. Color Contrast — Ensure enough contrast between text and background so low-vision users can read comfortably.

High vs. Low Contrast Example



The Practical Walkthrough: Adding Accessibility to a Web Page Mockup

Step 1: For every image on the page, write a short ALT text description of what it shows and why it matters (not just "image1.jpg").

Step 2: Structure your headings in order — Heading 1 for the page title, Heading 2 for major sections, Heading 3 for subsections — so a screen reader can announce the page's structure logically.

Step 3: Check text/background color combinations using a contrast-checking tool; aim for a ratio that meets accessibility guidelines (commonly a minimum of 4.5:1 for normal text).

Step 4: Test the page by closing your eyes and having someone read only the ALT text and headings aloud — see if the core message still comes through.

What just happened? You just made your content usable for a much wider audience — including users with visual impairments — using nothing but text descriptions and structure.

Grab a Partner

Partner A closes their eyes while Partner B reads a web page aloud using **only** the ALT text and headings provided (no visual description beyond what's written). Evaluate: would a visually impaired reader actually understand the page's core message from this alone? What's missing?

Quick Recap

Accessible design — readable fonts, meaningful ALT text, and strong color contrast — ensures digital content works for everyone, not just users without disabilities.

7.5.5.2 User Feedback

The Explanation

User feedback means gathering opinions from real users to improve digital content.

1. Surveys and Forms — Use surveys to collect feedback on digital content and make necessary improvements.

Example: After creating a website, ask users to share their experience and make adjustments based on their suggestions.

2. Direct Communication — Engage with users directly to understand their needs and preferences.

Example: Ask classmates for input on a presentation and use their feedback to improve it.

Quick Recap

Accessible, ethical design isn't a one-time task — ongoing user feedback ensures digital content keeps improving for the real people who actually use it.

Do You Know? — Pakistan's Data Protection Bill

In 2020, Pakistan introduced the **Personal Data Protection Bill**, which aims to protect citizens' data and ensure their privacy. This bill requires organizations to obtain explicit consent before collecting personal data and mandates the secure storage of such data.

Chapter Summary: The Big Picture

Let's connect everything we've learned into one digital citizenship journey:

1. KNOW YOUR RIGHTS → Terms of Use, personal rights, legal ramifications
2. RECOGNIZE THREATS → Spam, spyware, cookies, phishing, pharming
3. DEFEND YOURSELF → Spam filters, anti-malware, cookie management, URL checks
4. SEE THE BIGGER PICTURE → Digital divide, computing's societal impact
5. ACT ETHICALLY → Responsible sharing, avoiding plagiarism, respecting copyright
6. BUILD FOR EVERYONE → Accessible design, collaborative problem solving, user feedback

Every responsible digital citizen — whether a student, a software engineer, or a policymaker — moves through some version of this exact journey. You now understand every step of it.

Remember: digital ethics isn't a list of "don'ts." It's a toolkit for **empowerment**. Understanding your rights, recognizing manipulation, and designing with everyone in mind lets you protect yourself, protect your community, and help build a fairer digital future for the users who come after you.

Exercise

Multiple-Choice Questions (MCQs)

1. Option typically included in common Terms of Use clauses:
a) User obligations b) Privacy and data use c) Product advertising d) Termination of service
2. The purpose of the "Limitations of Liability" clause in Terms of Use:
a) Ensures full company responsibility b) Limits company's liability c) Provides user rights for damages d) Guarantees constant service availability
3. The type of harmful software that secretly monitors user activity:
a) Spam b) Cookies c) Spyware d) Pharming
4. The threat involving user redirection to fake websites:
a) Phishing b) Spam c) Spyware d) Pharming
5. The impact of the digital divide on education:
a) Improves learning for all b) No educational impact c) Creates inequality in digital resource access d) Replaces traditional teaching methods
6. The reason for understanding bias during information evaluation:
a) Ignoring other viewpoints b) Recognizing the source's agenda c) Validating all sources equally d) Avoiding evaluation altogether
7. The positive societal impact of computing systems:
a) Spread of misinformation b) Improved information accessibility c) Heightened privacy concerns d) Widened technology gap
8. The practice considered ethical when using information:
a) Copying content without permission b) Respecting copyright and avoiding plagiarism c) Ignoring source attribution d) Using unverified information
9. The behavior considered responsible digital conduct:
a) Using secure websites b) Spreading false info online c) Respecting online privacy d) Reporting suspicious activity

Short Questions

1. Why is it important for users to understand Terms of Use?
2. Differentiate between phishing and pharming.
3. Identify two impacts of the digital divide on social and civic participation.
4. What are the key steps involved in evaluating information sources?

5. How does responsible data sharing contribute to ethical use of information?

Long Questions

1. Explain the common clauses found in Terms of Use and describe how they protect both the service provider and the user.
2. Explain the concept of bridging the digital divide and discuss the roles of government initiatives in addressing this issue. Provide relevant examples from Pakistan.
3. Explain how critical thinking skills contribute to responsible information utilization. Provide examples of how these skills can be applied in real-life scenarios.
4. Discuss the importance of collaborative problem solving in a digital environment and provide examples of how human-machine collaboration can enhance this process.

End of Unit 7: Legal and Ethical Aspects of Computing System